



$$P = \frac{\Delta F}{\Delta A}$$

$$PV = NRT$$

$$\Delta V = \frac{\Delta x}{\Delta t}$$

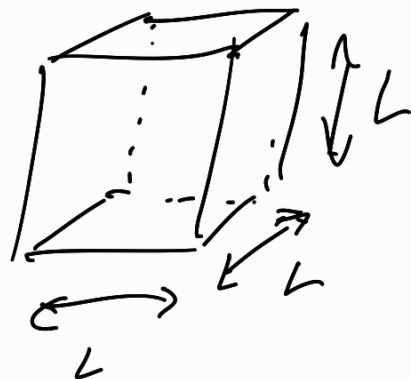
$$= \frac{l}{\Delta t} = 2V$$

$$\rightarrow \frac{2V}{l} = \frac{1}{\Delta t}$$

$$V - (-V) = 2V$$

$$F = ma = m \frac{\Delta V}{\Delta t} = m \frac{2V^2}{l}$$

$$\frac{F}{A} = \frac{m 2V^2}{lA} \approx P \leftarrow 1$$



$$LA = V$$

$$L L^2 =$$

$$\langle V^2 \rangle = \frac{V^2}{N}$$

$$P = \frac{m 2V^2}{3V}$$

$$PV = m 2V^2$$

$$\frac{1}{2} m \langle V^2 \rangle = \frac{f}{2} K T$$

$$\rightarrow \frac{3}{2} K T$$

$$\frac{PV}{N} = 3 K T \Rightarrow PV = 3 N K T$$

$$PV = N R T$$

$$\Delta K = \frac{f}{2} K T$$

$$\sim \frac{1}{2} m \langle V^2 \rangle = \frac{3}{2} K T$$

$$\langle V^2 \rangle = \frac{3}{m} K T$$

$$T = 90^\circ F = \frac{5}{9} T_c + 32 =$$

$$(58) \frac{5}{9} = T_c$$

$$\frac{360}{9} \approx \boxed{33.3 = T_c} + 273 \approx 300$$

$$\langle v^2 \rangle = \frac{3}{m} k T_c$$

$$m = N_2 = 28 m_p = 4.65 \times 10^{-27} \text{ kg}$$

$$k = 1.4 \times 10^{-23} \text{ m}^2 \text{ kg s}^{-2} \text{ K}^{-1}$$

$$\langle v^2 \rangle = \left(\frac{3}{4.65} \right) \times 10^{27} (1.4 \times 10^{-23}) (300)$$

$$= \left(\frac{3}{5} \right) \times 10^6$$

$$= \frac{9}{5} \times 10^6$$

$$\approx 2 \times 10^6 \left(\frac{\text{m}}{\text{s}} \right)^2$$

$$\sqrt{\langle v^2 \rangle} = \sqrt{2 \times 10^6}$$

$$\approx \sqrt{2} \times 10^3 \left(\frac{\text{m}}{\text{s}} \right)$$

$$\approx 1.4 \times 10^3 \frac{\text{m}}{\text{s}}$$

$$= 1400 \frac{\text{m}}{\text{s}}$$

$$\Rightarrow \frac{1400}{1600} \frac{\text{miles}}{\text{s}}$$

$$\frac{7}{8} \frac{\text{miles}}{\text{s}} \quad 3600$$

$$\approx 3200 \frac{\text{miles}}{\text{hr}}$$

$$\Delta P = \frac{\Delta F}{A} = \frac{Nmg}{A} = \frac{\Delta h \rho g N}{\Delta V}$$

$$V = hA \Rightarrow \frac{h}{V} = \frac{1}{A}$$

$$h \Delta \rho \left[\frac{1}{2} \rho \Delta x + \dots + \frac{1}{2} \rho \Delta x \right]$$

$$P = \frac{NkT}{\Delta V} = \frac{NkT}{P}$$

$$\Delta P = \frac{\Delta h \rho g P}{RT}$$

$$\int \frac{\Delta P}{P} = \int \Delta h \frac{mg}{kT} \quad \Delta h \rightarrow dh$$
$$\Delta P \rightarrow dP$$

$$\ln(P) = \frac{hmg}{kT}$$

$$P = P_0 e^{-hmg/kT}$$

Sea level
 P_0